

Project Report

BoardNet: Board Examination Communication Network

Base Network: 16.44.0.0/16

Derived from: Student ID $\Rightarrow$ 23201644 (Md. Rimon Hossain). The last four digits 1644 split $\Rightarrow$ 16 and 44

Contents:

1. VLSM Tree:

Table

Name	Hosts	Hosts + 2	Block Size	Host bits	Prefix
DHK	300	302	512	9	23
CTG	200	202	256	8	24
RAJ	160	162	256	8	24
SYL	150	152	256	8	24
BAR	120	122	128	7	25
KHU	120	122	128	7	25
DHK-SYL-RAJ	3	5	8	3	29
DHK-CTG	2	4	4	2	30
DHK-KHU	2	4	4	2	30
KHU-BAR	2	4	4	2	30
BAR-DHK	2	4	4	2	30

DHK-SYL-RAJ $\Rightarrow$ Central Switch

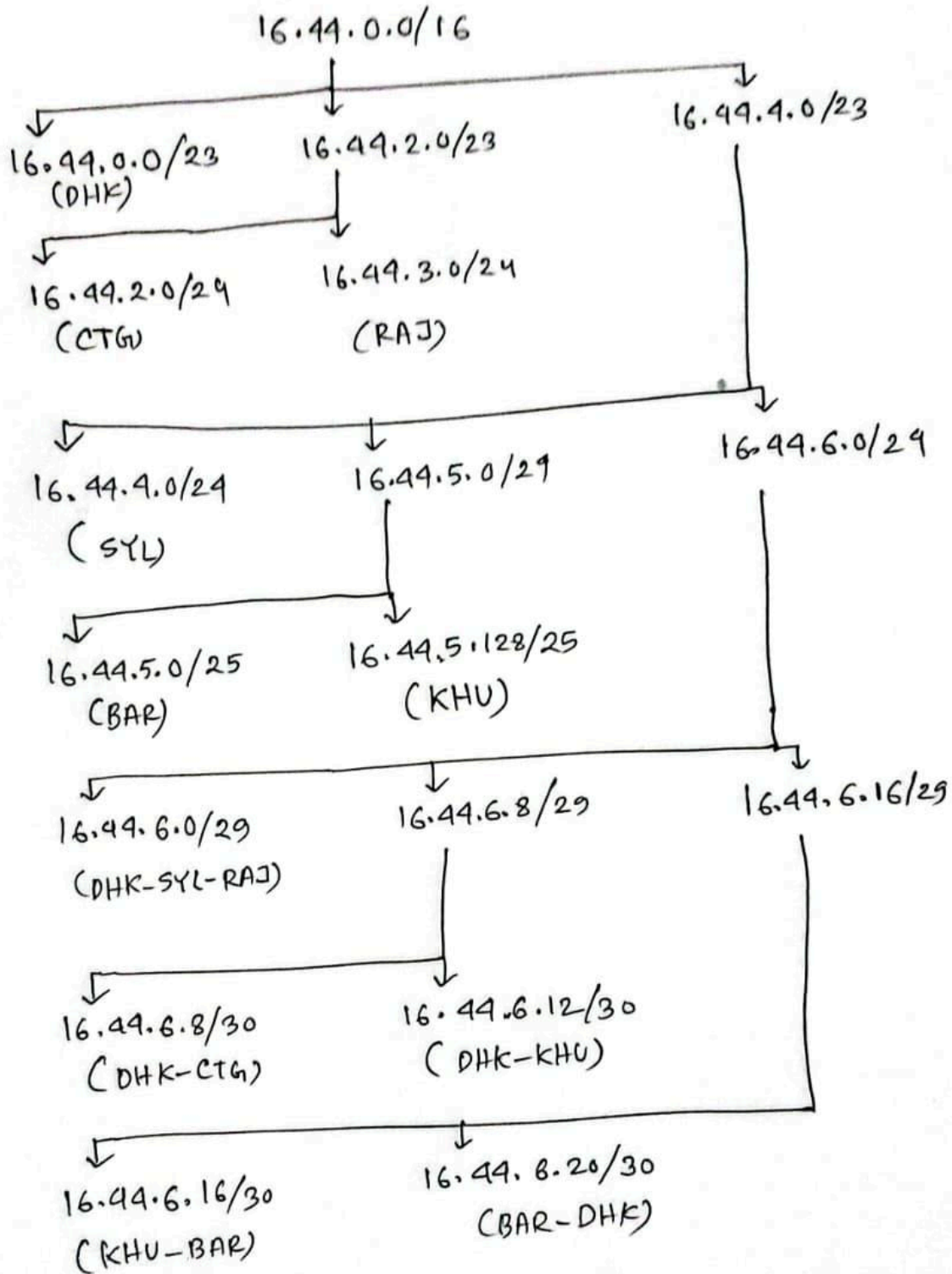
DHK-CTG
DHK-KHU
KHU-BAR
BAR-DHK

} WAN

$$\frac{16 \cdot 44 \cdot 0 \cdot 0}{8 \cdot 8 \cdot 8 \cdot 8} / 16$$

0000 0000	→ 0.0 (DHK)
0000 0000	→ 2.0 (CTG)
0000 0000	→ 3.0 (RAJ)
0000 0000	→ 4.0 (SYL)
0000 0000	→ 5.0 (BAR)
1000 0000	→ 5.128 (KHU)
0000 0000	→ 6.0 (DHK-SYL-RAJ)
0000 0000	→ 6.8 (DHK-CTG)
0000 0000	→ 6.12 (DHK-KHU)
0000 0000	→ 6.16 (KHU-BAR)
0000 0000	→ 6.20 (BAR-DHK)

VLSM Tree



The Full Table:

Subnet Name	Hosts Available	Unused Hosts	Network Address	Subnet Mask	Usable IP Range	Broadcast Address
1. DHK	510	210	16.44.0.0/23	255.255.254.0	16.44.0.1 - 16.44.1.254	16.44.1.255
2. CTG	254	54	16.44.2.0/24	255.255.255.0	16.44.2.1 - 16.44.2.254	16.44.2.255
3. RAJ	254	94	16.44.3.0/24	255.255.255.0	16.44.3.1 - 16.44.3.254	16.44.3.255
4. SYL	254	104	16.44.4.0/24	255.255.255.0	16.44.4.1 - 16.44.4.254	16.44.4.255
5. BAR	126	6	16.44.5.0/25	255.255.255.128	16.44.5.1 - 16.44.5.126	16.44.5.127

6. KHU	126	6	16.44.5.128/25	255.255.255.128	16.44.5.129 - 16.44.5.254	16.44.5.255
7. Central Switch (DHK-SYL-RAJ)	6	3	16.44.6.0/29	255.255.255.248	16.44.6.1 - 16.44.6.6	16.44.6.7
8. Link: DHK to CTG	2	0	16.44.6.8/30	255.255.255.252	16.44.6.9 - 16.44.6.10	16.44.6.11
9. Link: DHK to KHU	2	0	16.44.6.12/30	255.255.255.252	16.44.6.13 - 16.44.6.14	16.44.6.15
10. Link: KHU to BAR	2	0	16.44.6.16/30	255.255.255.252	16.44.6.17 - 16.44.6.18	16.44.6.19
11. Link: BAR to DHK	2	0	16.44.6.20/30	255.255.255.252	16.44.6.21 - 16.44.6.22	16.44.6.23

Full IP Addressing Table

Name	Device Name	Interface	IP Address	Subnet Mask	Default Gateway
DHK	Router	Fa1/0, LAN	16.44.0.1	255.255.254.0	—
DHK	Router	Fa0/0, Central Switch	16.44.6.1	255.255.255.248	—
DHK	Router	Se6/0, link to CTG	16.44.6.9	255.255.255.252	—
DHK	Router	Se2/0, link to KHU	16.44.6.13	255.255.255.252	—
DHK	Router	Se3/0, link to BAR	16.44.6.21	255.255.255.252	—
DHK	PC0	—	16.44.0.4	255.255.254.0	16.44.0.1
DHK	PC1	—	16.44.0.3	255.255.254.0	16.44.0.1
DHK	Printer	—	16.44.0.250	255.255.254.0	16.44.0.1
DHK	Central DNS Server	—	16.44.0.2	255.255.254.0	16.44.0.1
DHK	Mail Server	mail.dhk.edu.bd	16.44.0.50	255.255.254.0	16.44.0.1
DHK	Web Server	www.dhk.edu.bd	16.44.0.100	255.255.254.0	16.44.0.1
CTG	Router	Fa0/0, LAN	16.44.2.1	255.255.255.0	—
CTG	Router	Se2/0, link to DHK	16.44.6.10	255.255.255.252	—
CTG	PC2	—	16.44.2.2	255.255.255.0	16.44.2.1
CTG	PC3	—	16.44.2.3	255.255.255.0	16.44.2.1
CTG	Printer	—	16.44.2.250	255.255.255.0	16.44.2.1

CTG	Mail Server	mail.ctg.edu.bd	16.44.2.50	255.255.255.0	16.44.2.1
CTG	Web Server	www.ctg.edu.bd	16.44.2.200	255.255.255.0	16.44.2.1
RAJ	Router	Fa1/0, LAN	16.44.3.1	255.255.255.0	—
RAJ	Router	Fa0/0, Central Switch	16.44.6.3	255.255.255.248	—
RAJ	PC4	—	16.44.3.3	255.255.255.0	16.44.3.1
RAJ	PC5	—	16.44.3.2	255.255.255.0	16.44.3.1
RAJ	Printer	—	16.44.3.250	255.255.255.0	16.44.3.1
RAJ	Mail Server	mail.raj.edu.bd	16.44.3.100	255.255.255.0	16.44.3.1
SYL	Router	Fa1/0, LAN	16.44.4.1	255.255.255.0	—
SYL	Router	Fa0/0, Central Switch	16.44.6.2	255.255.255.248	—
SYL	PC6	—	16.44.4.2	255.255.255.0	16.44.4.1
SYL	PC7	—	16.44.4.3	255.255.255.0	16.44.4.1
SYL	Printer	—	16.44.4.250	255.255.255.0	16.44.4.1
SYL	Mail Server	mail.syl.edu.bd	16.44.4.50	255.255.255.0	16.44.4.1
BAR	Router	Fa0/0, LAN	16.44.5.1	255.255.255.128	—
BAR	Router	Se2/0, link to KHU	16.44.6.18	255.255.255.252	—
BAR	Router	Se3/0, link to DHK	16.44.6.22	255.255.255.252	—
BAR	PC8	—	16.44.5.11	255.255.255.128	16.44.5.1
BAR	PC9	—	16.44.5.10	255.255.255.128	16.44.5.1

BAR	Printer	—	16.44.5.125	255.255.255.128	16.44.5.1
BAR	Mail Server	mail.bar.edu.bd	16.44.5.50	255.255.255.128	16.44.5.1
BAR	Local DHCP Server	—	16.44.5.2	255.255.255.128	16.44.5.1
KHU	Router	Fa0/0, LAN	16.44.5.129	255.255.255.128	—
KHU	Router	Se2/0, link to DHK	16.44.6.14	255.255.255.252	—
KHU	Router	Se3/0, link to BAR	16.44.6.17	255.255.255.252	—
KHU	PC10	—	16.44.5.141	255.255.255.128	16.44.5.129
KHU	PC11	—	16.44.5.140	255.255.255.128	16.44.5.129
KHU	Printer	—	16.44.5.250	255.255.255.128	16.44.5.129
KHU	Mail Server	mail.khu.edu.bd	16.44.5.150	255.255.255.128	16.44.5.129
KHU	Local DHCP Server	—	16.44.5.130	255.255.255.128	16.44.5.129

Router Configuration Commands

DHK Router (Core RIP & DHCP Server)

```
enable
configure terminal
interface Fa0/0
ip address 16.44.0.1 255.255.254.0
no shutdown
exit
```

```
ip dhcp pool CTG_POOL
network 16.44.2.0 255.255.255.0
default-router 16.44.2.1
dns-server 16.44.0.2
exit
```

```
router rip
version 2
network 16.44.0.0
network 16.44.6.0
redistribute static
exit
```

CTG Router

```
enable
configure terminal
interface Fa0/0
ip address 16.44.2.1 255.255.255.0
ip helper-address 16.44.6.9
no shutdown
```

exit

ip route 0.0.0.0 0.0.0.0 16.44.6.9
exit

RAJ Router

enable
configure terminal
interface Fa0/0
ip address 16.44.3.1 255.255.255.0
ip helper-address 16.44.6.1
no shutdown
exit

router rip
version 2
network 16.44.3.0
network 16.44.6.0
exit

SYL Router

enable
configure terminal
interface Fa0/0
ip address 16.44.4.1 255.255.255.0
ip helper-address 16.44.6.1
no shutdown
exit

router rip
version 2
network 16.44.4.0
network 16.44.6.0
exit

BAR Router

enable

```
configure terminal
interface Fa0/0
ip address 16.44.5.1 255.255.255.128
no shutdown
exit
```

```
interface Se3/0
shutdown
! (Network reroutes via KHU)
no shutdown
exit
```

KHU Router

```
enable
configure terminal
interface Fa0/0
ip address 16.44.5.129 255.255.255.128
no shutdown
exit
```

```
interface Se2/0
shutdown
! (Network reroutes via BAR)
no shutdown
exit
```

Central DNS Server

Parameter	Configuration
Location	DHK
IPv4 Address	16.44.0.2
Subnet Mask	255.255.254.0
Default Gateway	16.44.0.1

DNS Service	ON
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DNS A Records

Domain Name	IP Address
www.dhk.edu.bd	16.44.0.100
www.ctg.edu.bd	16.44.2.200
mail.raj.edu.bd	16.44.3.100
mail.ctg.edu.bd	16.44.2.50
mail.syl.edu.bd	16.44.4.50
mail.bar.edu.bd	16.44.5.50
mail.khu.edu.bd	16.44.5.150
mail.dhk.edu.bd	16.44.0.50

DHCP Configuration:

The DHK Configuration Code:

Go to the **DHK** router's CLI and type these exact commands:

```
enable
configure terminal

ip dhcp excluded-address 16.44.0.1 16.44.0.2
ip dhcp excluded-address 16.44.2.1
ip dhcp excluded-address 16.44.3.1
ip dhcp excluded-address 16.44.4.1

ip dhcp pool DHK_LAN
network 16.44.0.0 255.255.254.0
```

```
default-router 16.44.0.1
dns-server 16.44.0.2
exit
```

```
ip dhcp pool CTG_LAN
network 16.44.2.0 255.255.255.0
default-router 16.44.2.1
dns-server 16.44.0.2
exit
```

```
ip dhcp pool RAJ_LAN
network 16.44.3.0 255.255.255.0
default-router 16.44.3.1
dns-server 16.44.0.2
exit
```

```
ip dhcp pool SYL_LAN
network 16.44.4.0 255.255.255.0
default-router 16.44.4.1
dns-server 16.44.0.2
exit
```

```
end
write memory
```

DHCP Relay (IP Helper) for CTG, SYL, and RAJ CTG Router:

```
enable
configure terminal
interface f0/0
ip helper-address 16.44.0.1
exit
write memory
```

SYL Router

```
enable
configure terminal
interface f1/0
ip helper-address 16.44.0.1
exit
```

```
end
write memory
memory
```

RAJ Router

```
enable  
configure terminal  
interface f1/0  
ip helper-address 16.44.0.1  
exit
```

```
end  
write memory
```

Local DHCP Servers for KHU and BAR

Static:

IPv4 Address: 16.44.5.2
Subnet Mask: 255.255.255.128
Default Gateway: 16.44.5.1
DNS Server: 16.44.0.2

DHCP Service

Default Gateway: 16.44.5.1
DNS Server: 16.44.0.2
Start IP Address: 16.44.5.10
Subnet Mask: 255.255.255.128

Khulna's Local DHCP Server

Static:

IPv4 Address: 16.44.5.130
Subnet Mask: 255.255.255.128
Default Gateway: 16.44.5.129
DNS Server: 16.44.0.2

DHCP Service

Default Gateway: 16.44.5.129
DNS Server: 16.44.0.2
Start IP Address: 16.44.5.140
Subnet Mask: 255.255.255.128

Web server configuration

DHK Web Server (www.dhk.edu.bd)

IPv4 Address: 16.44.0.200

Subnet Mask: 255.255.254.0

Default Gateway: 16.44.0.1

DNS Server: 16.44.0.2

CTG Web Server (www.ctg.edu.bd)

IPv4 Address: 16.44.2.200

Subnet Mask: 255.255.255.0

Default Gateway: 16.44.2.1
DNS Server: 16.44.0.2

Web Server: www.dhk.edu.bd (Located in DHK)

Parameter	Configuration
Location	DHK
IPv4 Address	16.44.0.100
Subnet Mask	255.255.254.0
Default Gateway	16.44.0.1
DNS Server	16.44.0.2
HTTP Service	ON

Web Server: www.ctg.edu.bd (Located in CTG)

Parameter	Configuration
Location	CTG
IPv4 Address	16.44.2.200
Subnet Mask	255.255.255.0
Default Gateway	16.44.2.1
DNS Server	16.44.0.2

HTTP Service	ON
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